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Sequence Bioinformatics **WS 2026/27**

Assignment 5 **Due: Wed 18-Nov-2026, 10:00**

This assignment has two parts. Part A applies the Bayesian inference of Monday's lecture (Chapter 4 of the handout, Sections 4.7 and 4.8) to the twelve primate sequences that run through Chapter 4, with the program BEAST 2, so that you can compare the posterior with the trees that distances and maximum likelihood gave in Sections 4.2.6 and 4.6.3. Part B is on Wednesday's lecture, the suffix tree (Chapter 11, Sections 11.2 and 11.3.1): you build the suffix tree of a short text by hand and use it to search. Section, figure, algorithm and exercise numbers refer to the handouts. Positions in a string are counted from 1, as in the handout.

Setup

Download `assignment05.zip` from Ilias. It contains `Primates.nex`, the alignment of twelve primate mitochondrial sequences of 898 sites (Hayasaka et al., 1988) that Chapter 4 uses, in NEXUS format, as distributed with BEAST. For Part A you need BEAST 2 (version 2.7, `https://www.beast2.org`; the package contains BEAUti, BEAST, TreeAnnotator and DensiTree), Tracer (`https://github.com/beast-dev/tracer/releases`) and a tree viewer that shows node labels (FigTree, Dendroscope, SplitsTree or the web tool IcyTree). All run on Linux, macOS and Windows. Part B needs pencil and paper.

Part A: Bayesian inference with BEAST 2

1 Setting up the analysis in BEAUti (2 points)

Open BEAUti and import `Primates.nex` (File > Import Alignment). Go through the tabs and set: on *Site Model*, the substitution model HKY (leave the other settings as they are); on *Clock Model*, Strict Clock; on *Priors*, the tree prior Yule Model (leave the priors of the parameters as they are); on *MCMC*, the chain length 200000, and keep the defaults for the logs (every 1000 generations). Save the configuration as `primates-short.xml`.

(a) Bayes' theorem, Section 4.7, writes the posterior as $P(T \mid A) \propto P(A \mid T)\,P(T)$. Which of the settings on the four tabs enter the likelihood $P(A \mid T)$, and which the prior $P(T)$? One sentence per tab.

(b) The Yule model is the Yule–Harding prior of Section 4.7. What does it assume about how trees grow, and which tree shapes does it favor, compared with the uniform prior?

(c) What does the chain length mean, and how many trees will the run write to the tree log?

2 Running BEAST and inspecting the run in Tracer (2 points)

Run BEAST on `primates-short.xml`. How long did the run take, which files did it write, and what do they contain? Then load `primates-short.log` into Tracer. The *effective sample size* (ESS) of a quantity is the number of independent draws that would carry as much information as the correlated

samples of the chain do; Tracer estimates it from the autocorrelation of the trace, and marks values below 200 in yellow and below 100 in red.

(a) Report the ESS of `posterior`, `likelihood`, `prior`, `TreeHeight` and `kappa`. Which are below 200, and what does a low ESS say about the run?

(b) Look at the trace of the posterior (the Trace tab). Where does the burn-in end, and why is it discarded (Section 4.7)? Why are the values of the likelihood and of the posterior negative?

(c) Report the mean and the 95% HPD interval of `kappa`, the transition/transversion rate ratio of the HKY model (Section 4.5.4), and of `TreeHeight`. Would you trust these estimates? Why, or why not?

3 A longer chain (2 points)

Return to BEAUti, set the chain length to 2000000, save as `primates-long.xml` and run BEAST again. Compare with the short run: the running time, the ESS values, the traces, and the estimates of `kappa` and `TreeHeight`. Why does a chain ten times as long give about ten times the ESS, while the estimates themselves move little? From your short run, how long a chain would you need to get every ESS above 200?

4 Summarizing the sample (2 points)

(a) Run TreeAnnotator on `primates-long.trees` with a burn-in of 10%, node heights "median heights" and the target tree type "Maximum clade credibility tree". Open the result in a tree viewer and display the posterior probability of each node (the annotation `posterior`). Is the topology the one that Neighbor-Joining and maximum likelihood found (Figure 4.5)? Which clades have posterior probability 1.0, which has the lowest, and what is uncertain there? Compare the posterior probability of the clade of human and chimpanzee with its bootstrap support in Table 4.1 (56% under Jukes–Cantor, 93% under the selected model): what do bootstrap values and posterior clade probabilities have in common, and in what do they differ (Section 4.8)?

(b) Open `primates-long.trees` in DensiTree. Describe the picture and compare it with Figure 4.12 of the handout: where is the bundle of trees tight and where does it spread, and which kind of uncertainty does the spread show? Include a screenshot of the tree of (a) and one of the DensiTree plot.

Part B: The suffix tree

5 A suffix tree by hand (2 points)

Let $T =$ `tgcacgca$`, with $n = 9$.

(a) Build the suffix tree of T by the naive construction, Algorithm 11.2, inserting the suffixes $T_1, T_2, \ldots, T_9$ in this order. Draw the tree after the insertion of T_4, after T_5, after T_7 and after T_9 (the final tree), with every edge label written out and every leaf labeled with the start position of its suffix. For each of the insertions T_5 to T_9, state in one line how many letters the walk from the root matches, whether it ends at an existing node or splits an edge, and which node and leaf edge are added. How many internal nodes besides the root does the final tree have, and which repeats of T are they?

(b) Use the final tree to find all occurrences in T of the queries `gca`, `cac`, `cat` and `gcg`, by Algorithm 11.3. For each query say where the walk from the root ends (at which node, or inside which edge after how many letters, or with which mismatch), how many letters of the query were compared, and which positions are reported. Check the positions against the text.